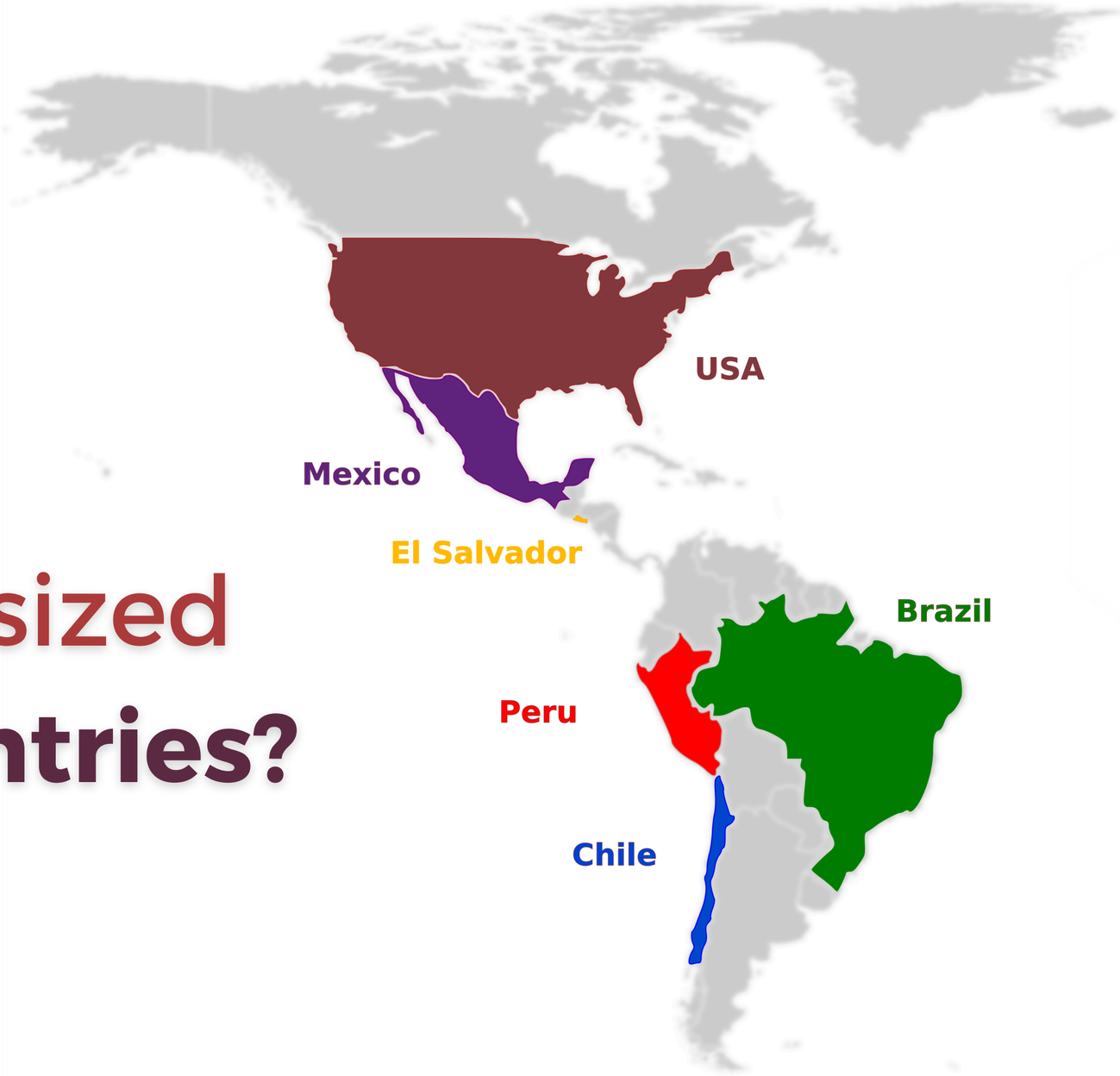
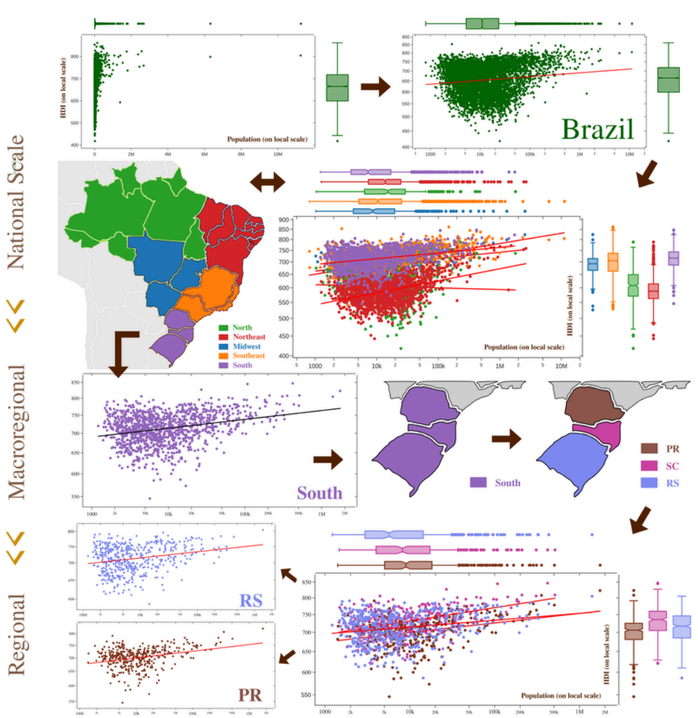
How to compare small and medium-sized cities between countries?

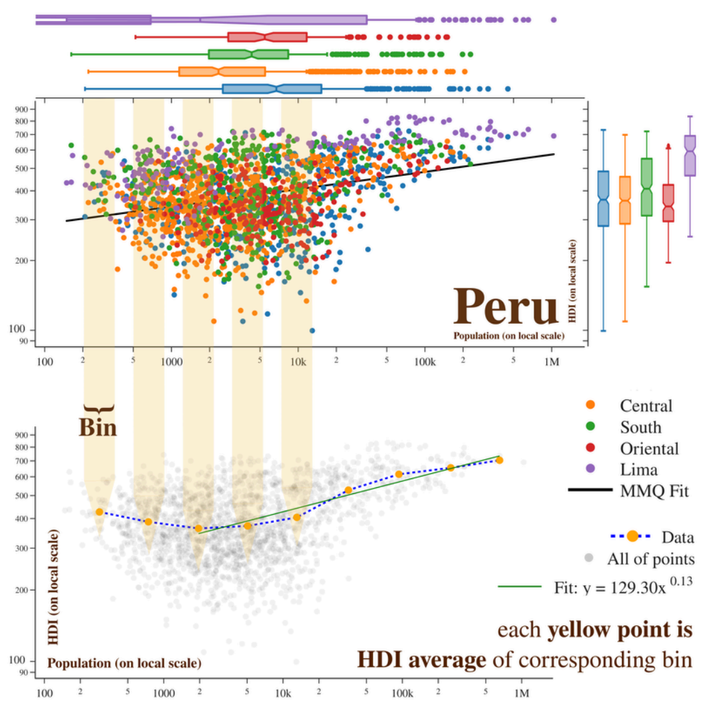


How to compare indicators between countries ?

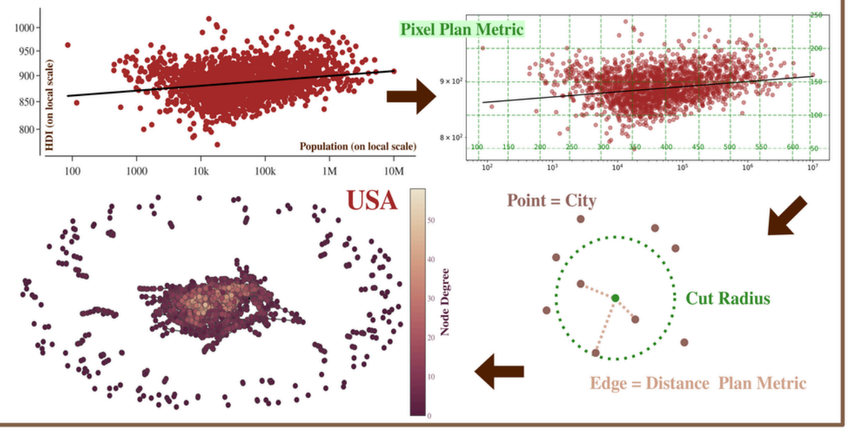
1) Detecting Scale-Free Pattern



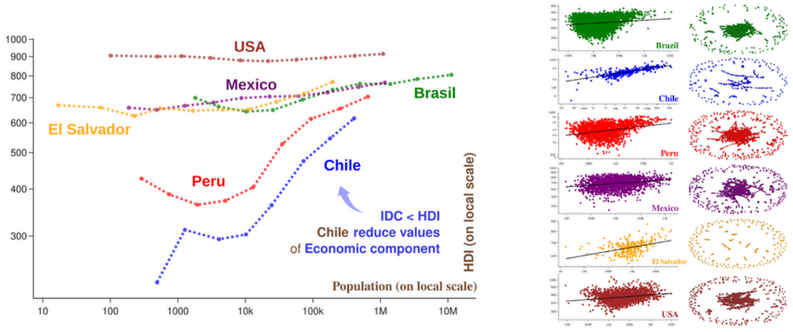
2) Binscatter Process



3) Based On GCA (Gravitational Clustering Algorithm)



4) Some Compared Results



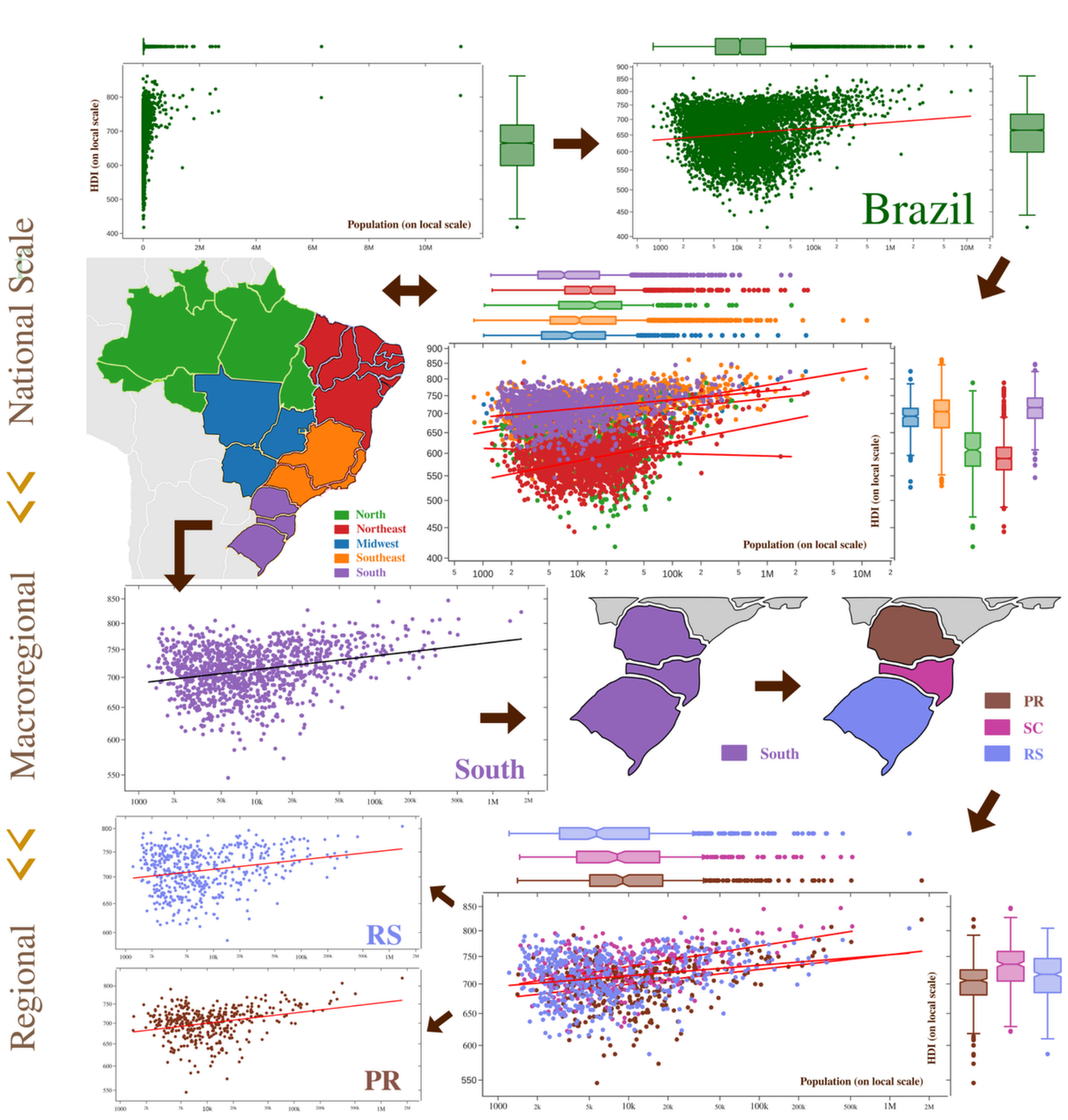
5) Conclusion: Multidisciplinary methodologies using visual mathematics and scale-free paterns can serve as the basis for definitive solutions in the future.

L. C. Rossato, J. R. Iglesias, and S. Gonçalves

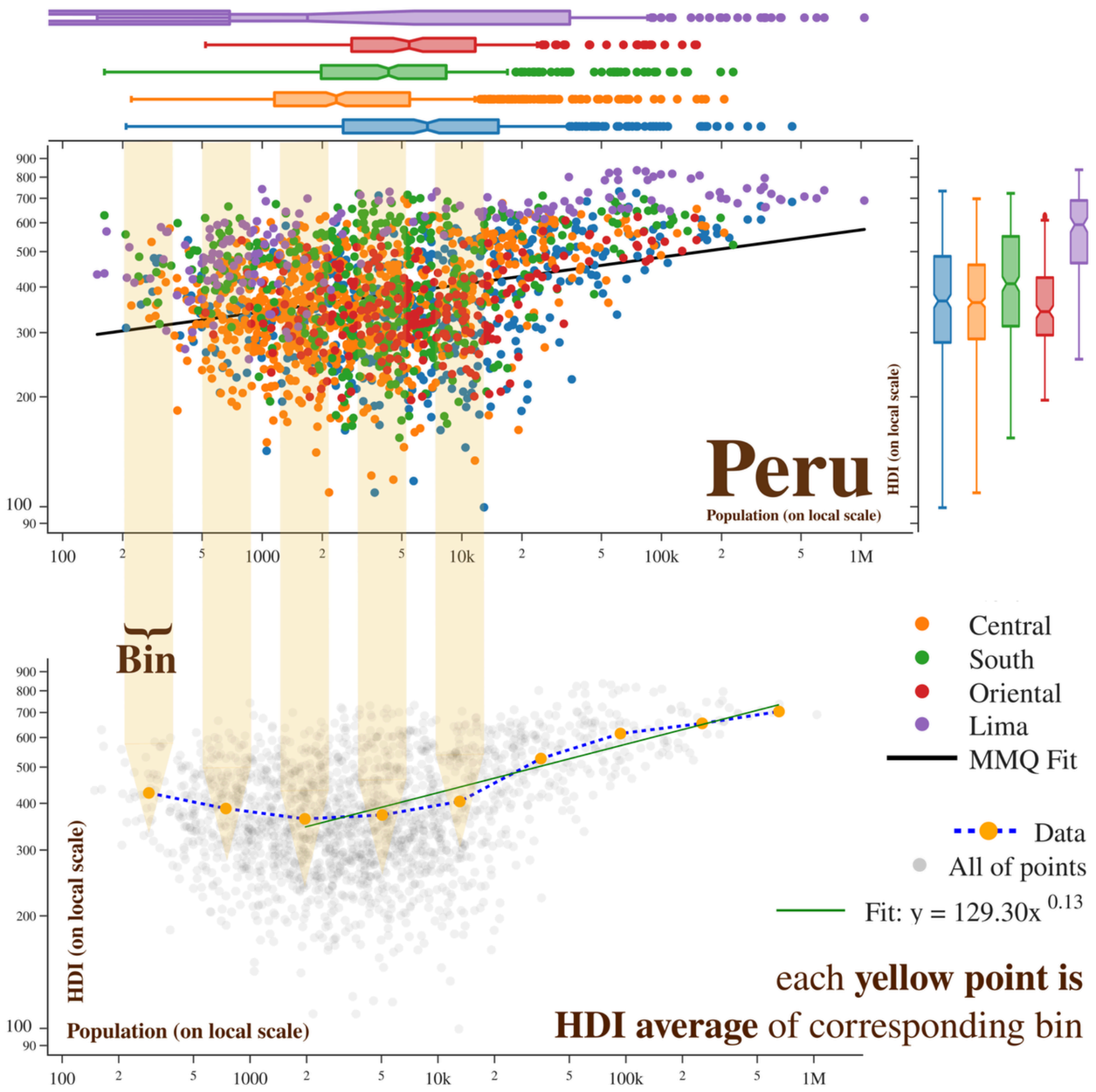
Instituto de Física

UFRGS
UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL

1) Detecting Scale-Free Pattern



2) Binscatter Process



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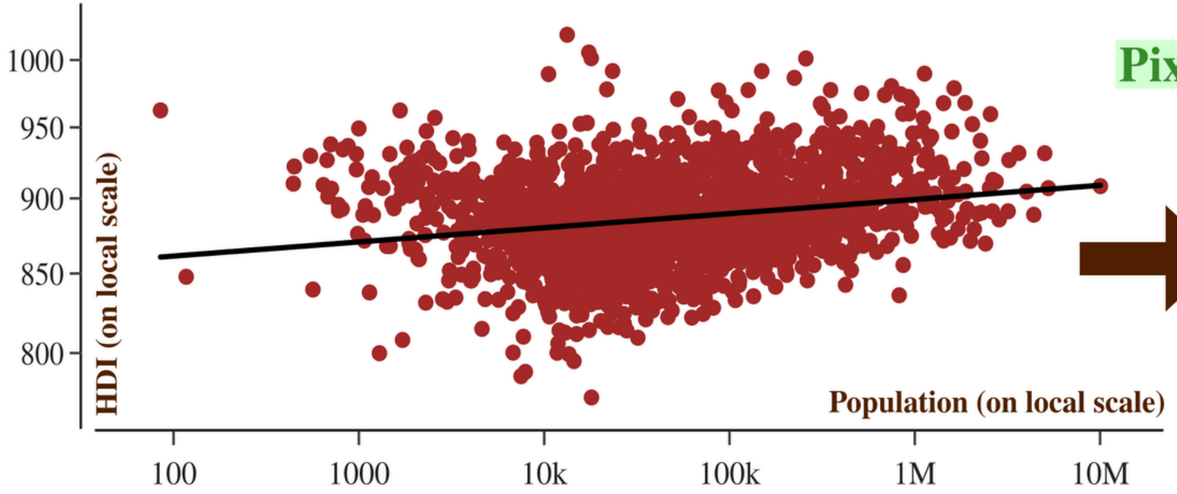
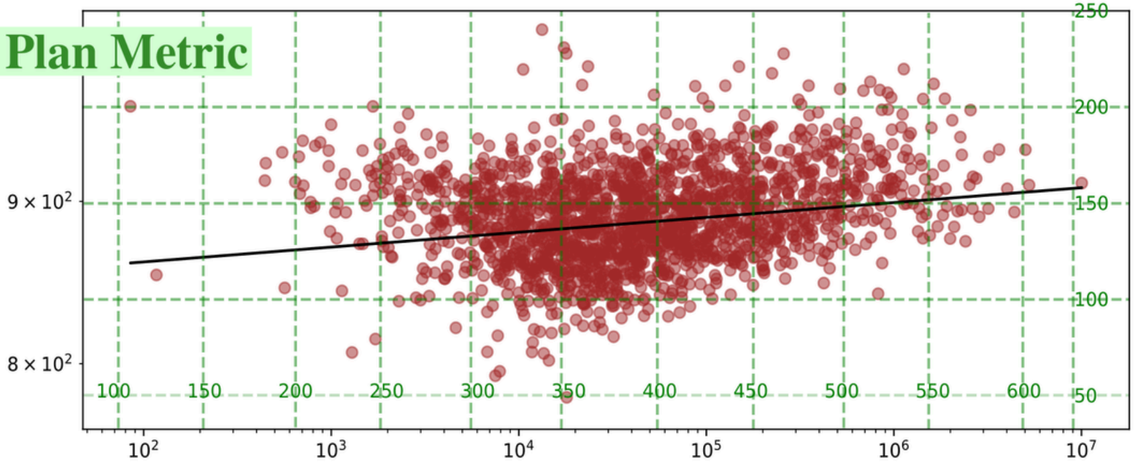
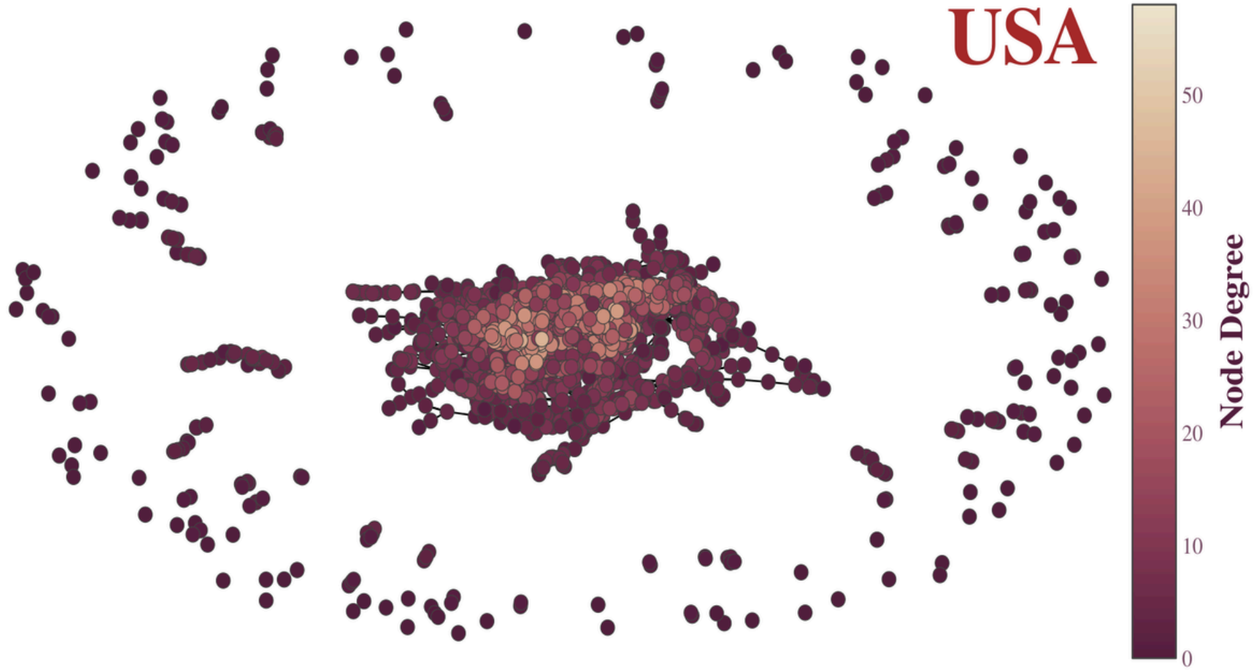
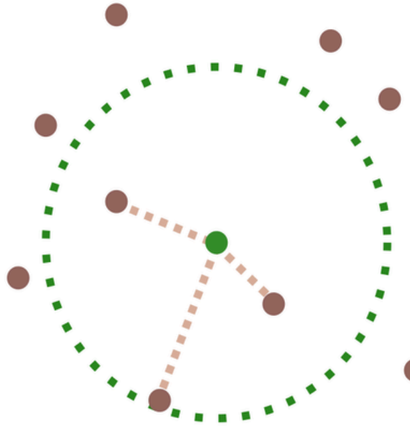


Pixel Plan Metric





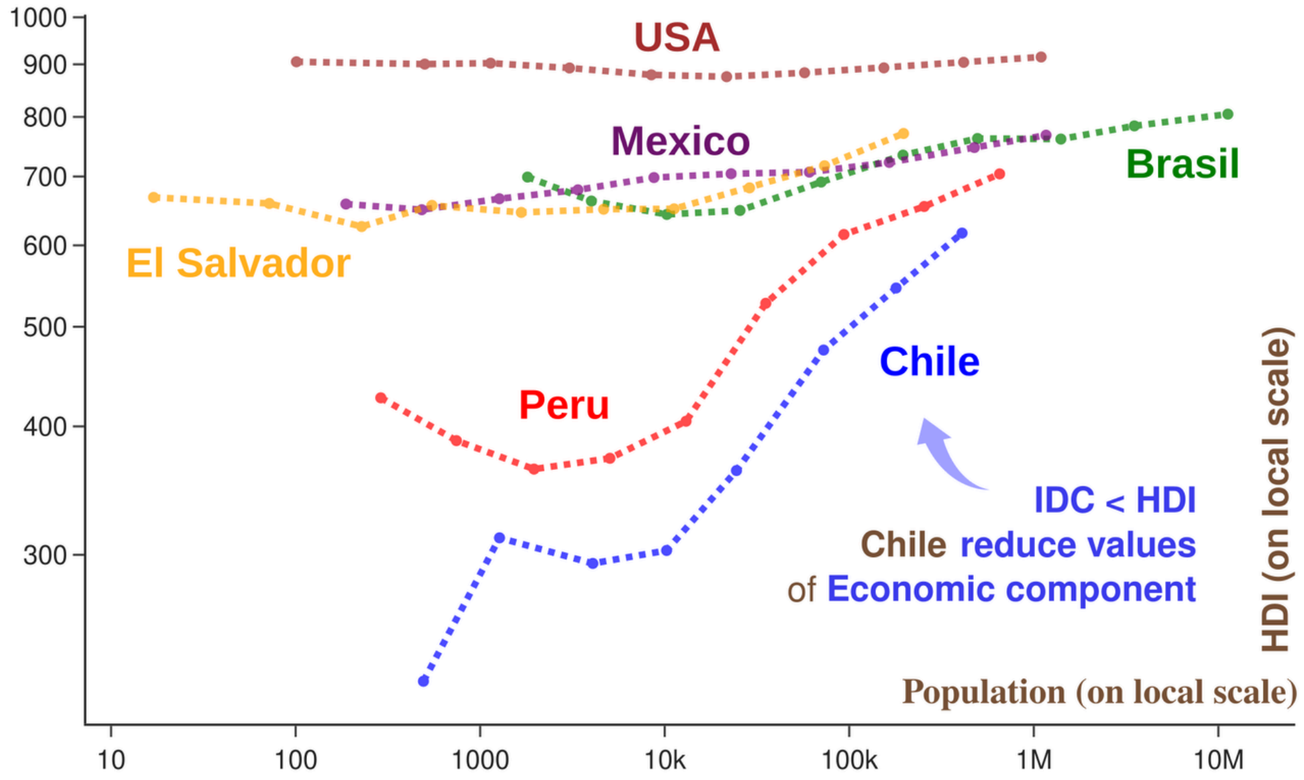
Point = City

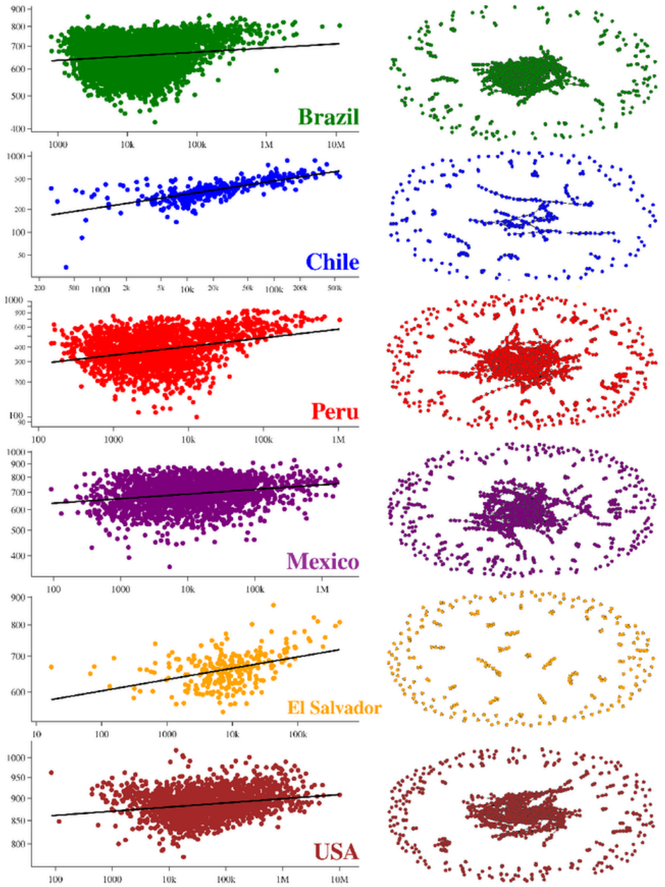


Cut Radius

Edge = Distance Plan Metric

4) Some Compared Results





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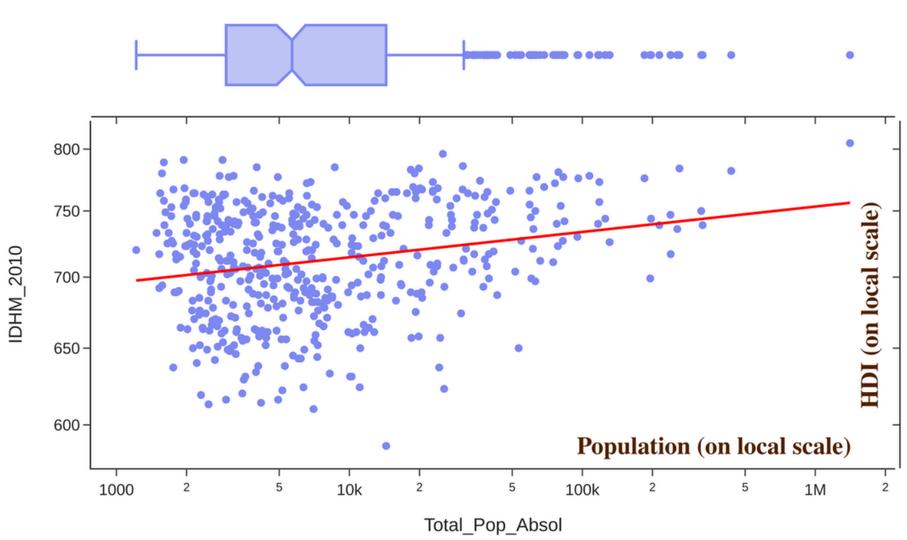
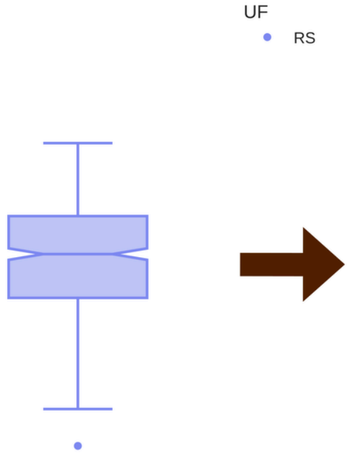
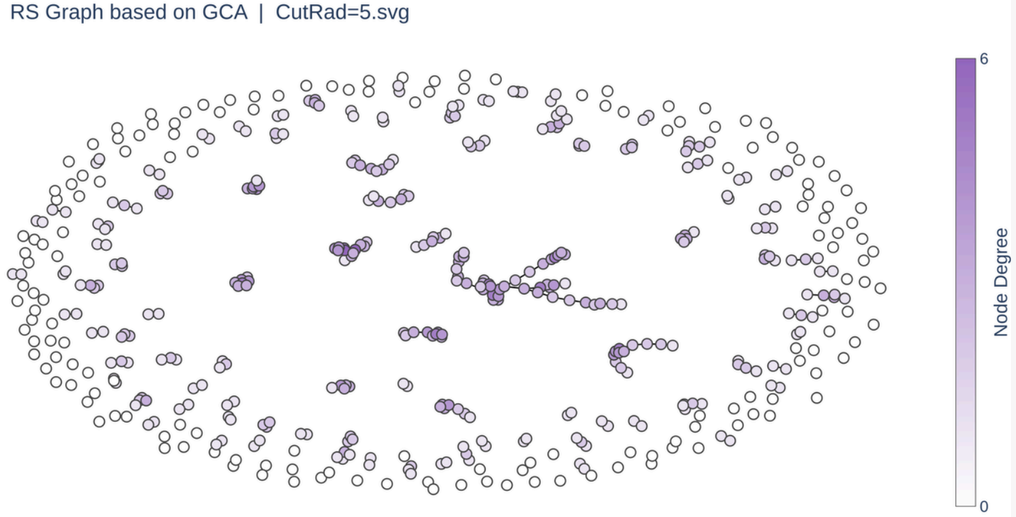
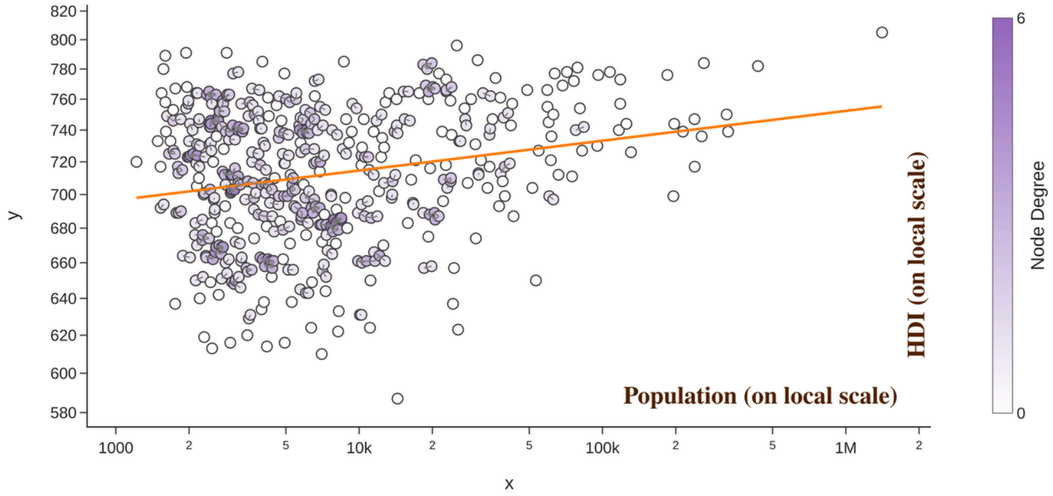






RS - Scatter Graph



How to compare small and medium-sized cities between countries?

Solutions is a web tool, make in Streamlit App: [City Comparison]

Analysis with Econophysics Techniques

NormalScatters Visualizing | BinScatters Visualizing | ScatterGraph Visualizing | DataFrame Analysis

Main Dashboard: Comparing Cities, Countries and Macrorregions

Comparing Cities | Comparing Countries | Comparing Macrorregions

Choose 2 cities for compare

Deploy

Main Dashboard: Cities Compare

Choose 2 cities for compare

City 1:

Select Country

Brazil

Select State

RS

Select City

São Luiz Gonzaga (RS)

City 2:

Select Country

Peru

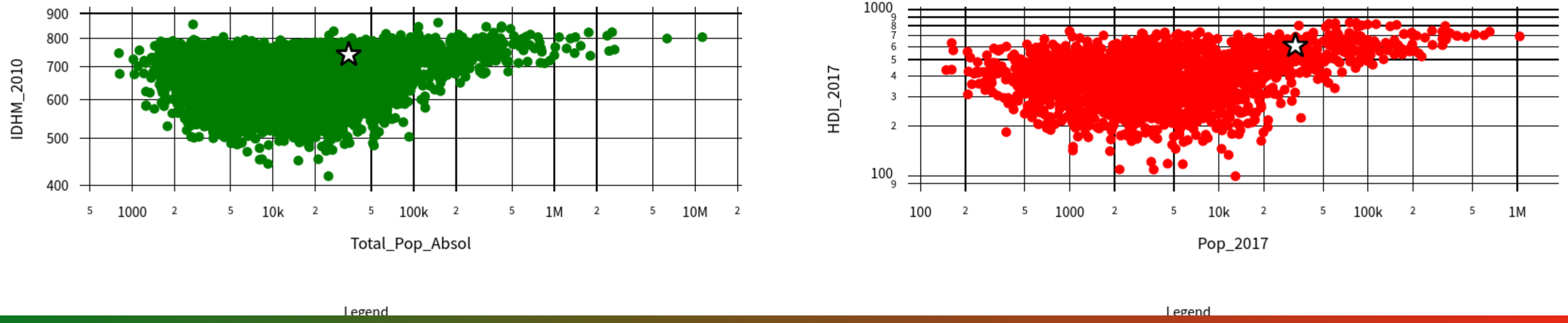
Select State

AMAZONAS

Select City

Chachapoyas

Brazil x **Peru**



USA

Back to Main Dashboard

Select Country

USA

Show DataFrame

Data File: `tabelas_csv/df_USA_Per_County.csv`

Settings

Select X-axis

2015_Tract_population

Select Y-axis

unadjusted_hdi

Select Color Column

Macroregion

Select Hovername Column

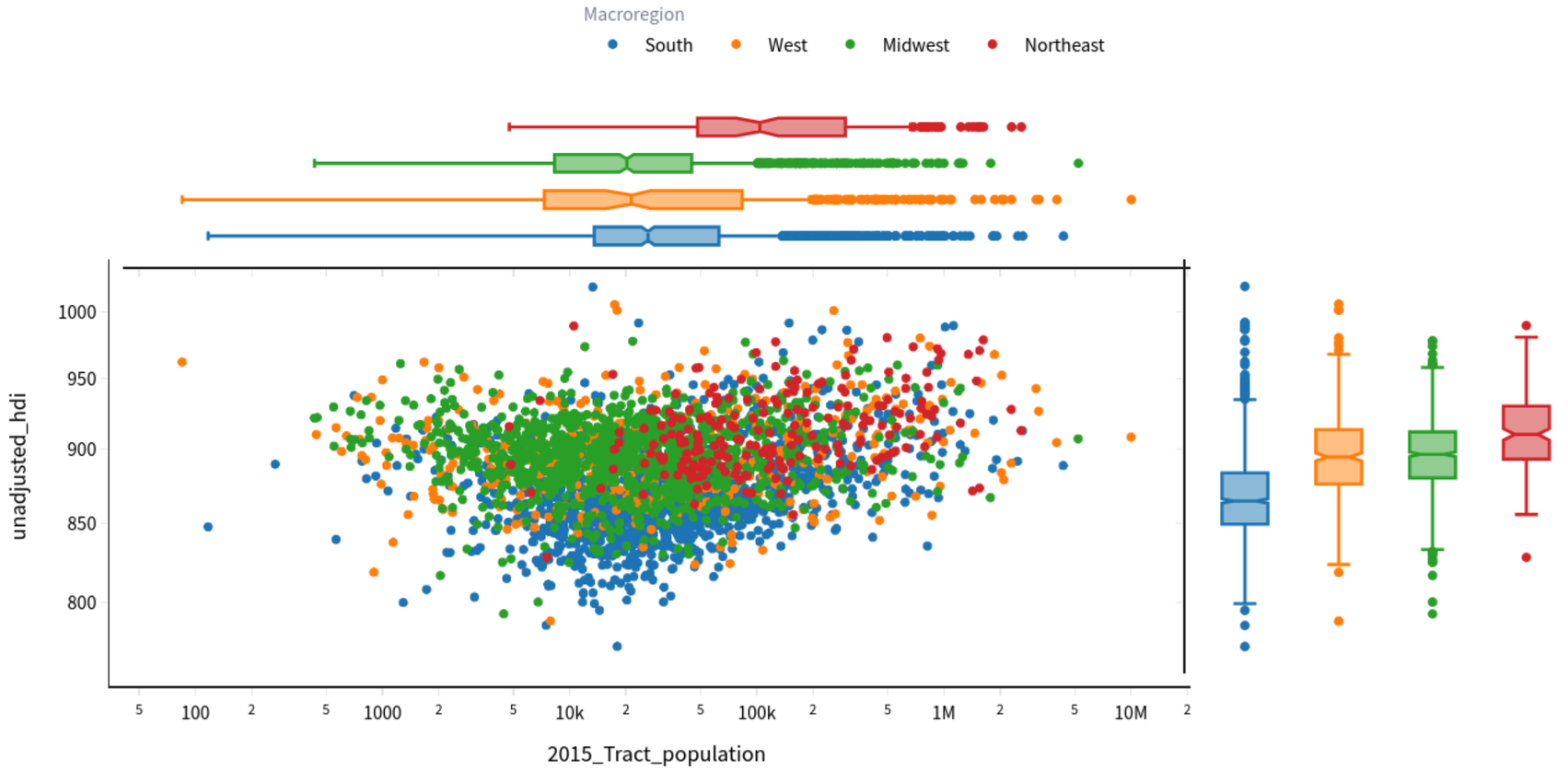
County_States

Deploy

Per Country | Per Macrorregion | Per Regions / States

Visualization Options:

Scatter Complete (with colors) | Scatter without Regions | Scatter without Trendline | Simple Scatter (Normal Scale)



México

Back to Main Dashboard

Select Country: Mexico

Show DataFrame

Data File: `tabelas_csv/df_Mexico.csv`

Settings

Select X-axis: Pop_Total_2010

Select Y-axis: IDH_2000

Select Color Column: Macroregioes

Select Hovername Column: Hovername

Deploy

1 — 1 — 100 — exponential

Data count by bin:

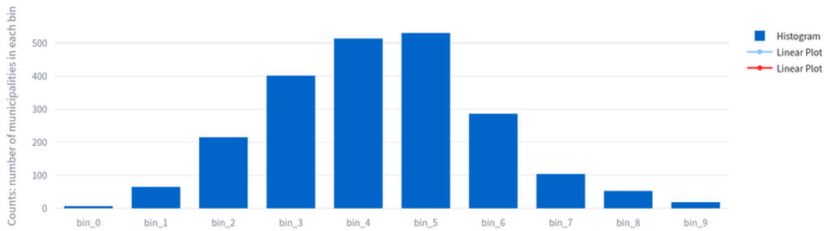
	bin_0	bin_1	bin_2	bin_3	bin_4	bin_5	bin_6	bin_7	bin_8	bin_9
count	7	65	215	401	513	530	286	104	53	19

Nº of Bins = 10

Bin averages:

	0	1	2	3	4	5	6	7	8	9
Pop_Total_2010	187.29	482.49	1,270.73	3,387.78	8,744.05	22,885.95	60,767.16	164,759.22	476,274.68	1,164,977.42
IDH_2000	658.14	649.8	666.01	679.38	698.33	704.34	706.95	722.47	746.94	767.89

Analysis IDH_2000 x Pop_Total_2010 by bins



Back to Main Dashboard

Select Country: Mexico

Show DataFrame

Data File: `tabelas_csv/df_Mexico.csv`

Settings

Select X-axis: Pop_Total_2010

Select Y-axis: IDH_2000

Select Color Column: Macroregioes

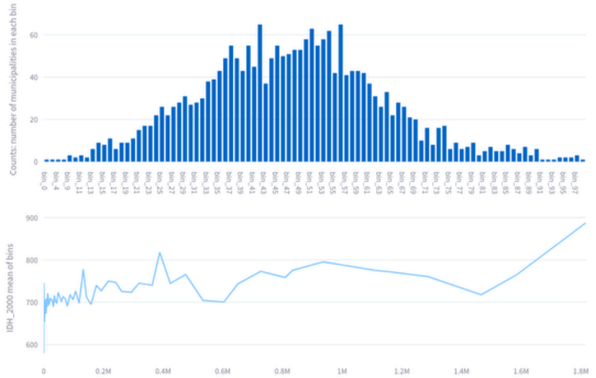
Select Hovername Column: Hovername

	bin_0	bin_1	bin_2	bin_3	bin_4	bin_5	bin_6	bin_7	bin_8	bin_9	bin_10	bin_11	bin_12	bin_13	bin_14	bin_15	bin_16	bin_17	bin_18	bin_19	bin_20	bin_21	bin_22	bin_23	bin_24	bin_
count	1	0	0	1	1	0	0	0	1	3	2	3	2	6	9	8	11	6	9	9	11	15	17	17	22	

Bin averages:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Pop_Total_2010	93	None	None	131	153	None	None	None	219	238.33	264.5	299.67	327	362.5	399.33	444.5	488	530.5	601.33	665.78	730.18	798.87
IDH_2000	719	None	None	580	648	None	None	None	735	641.67	654	745.67	585.5	602.33	684.22	645.38	634.91	675	647.89	635.67	625.36	670.33

Analysis IDH_2000 x Pop_Total_2010 by bins

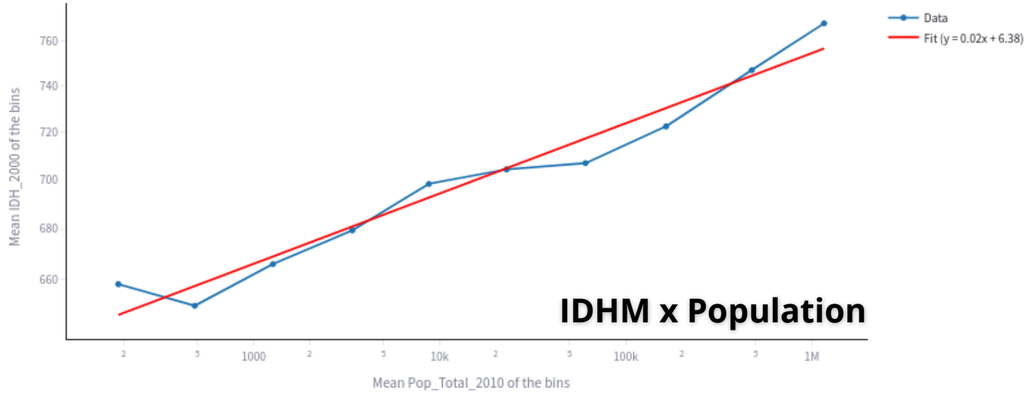


IDHM Distribution

Nº of Bins = 100

Binscatter: Fit Curve per Binscatter

Binscatter: Fit Curve Mexico

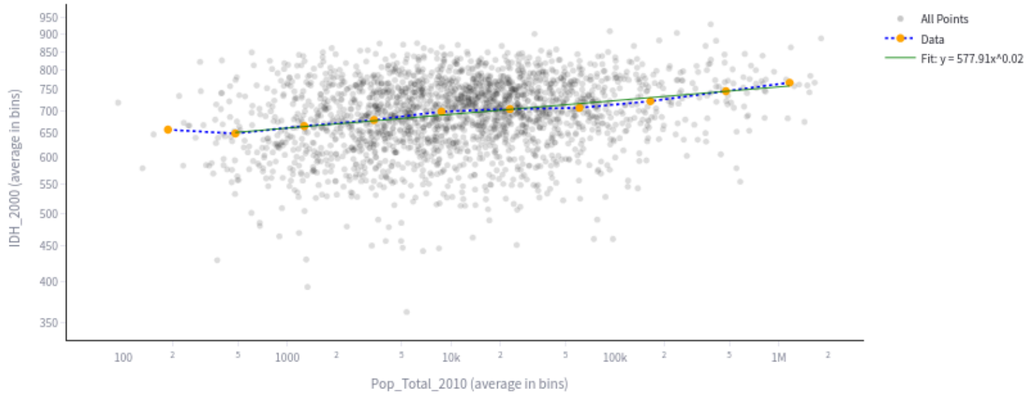


IDHM x Population

Binscatter vs Scatter

Select Opacity (Cities Points - background)

0.01 — 0.12



Back to Main Dashboard

Select Country

Peru

Show DataFrame

Data File: `tabelas_csv/df_Peru.csv`

Settings

Select X-axis

Pop_2017

Select Y-axis

HDI_2017

Select Color Column

Macroregioes_Topografica

Select Hovername Column

City

Visualization Options:

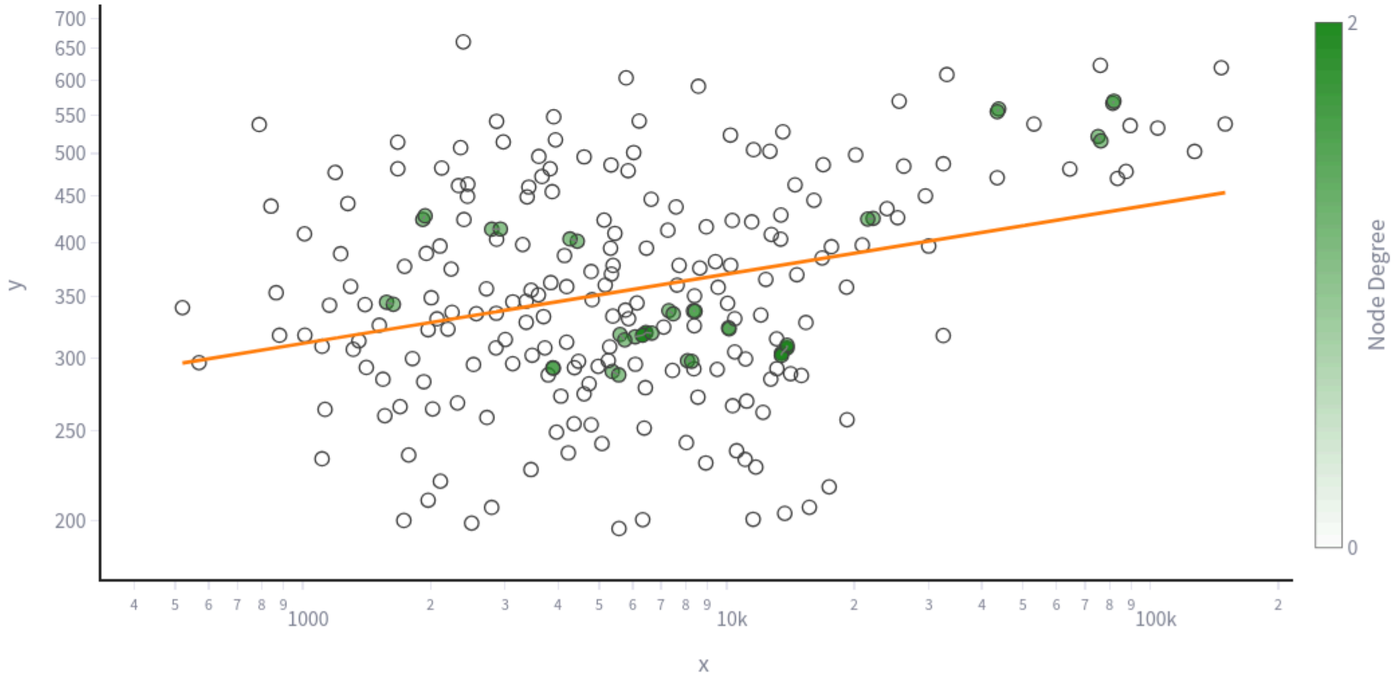
ScatterGraph | BGCA (Graph) | NormalScatter

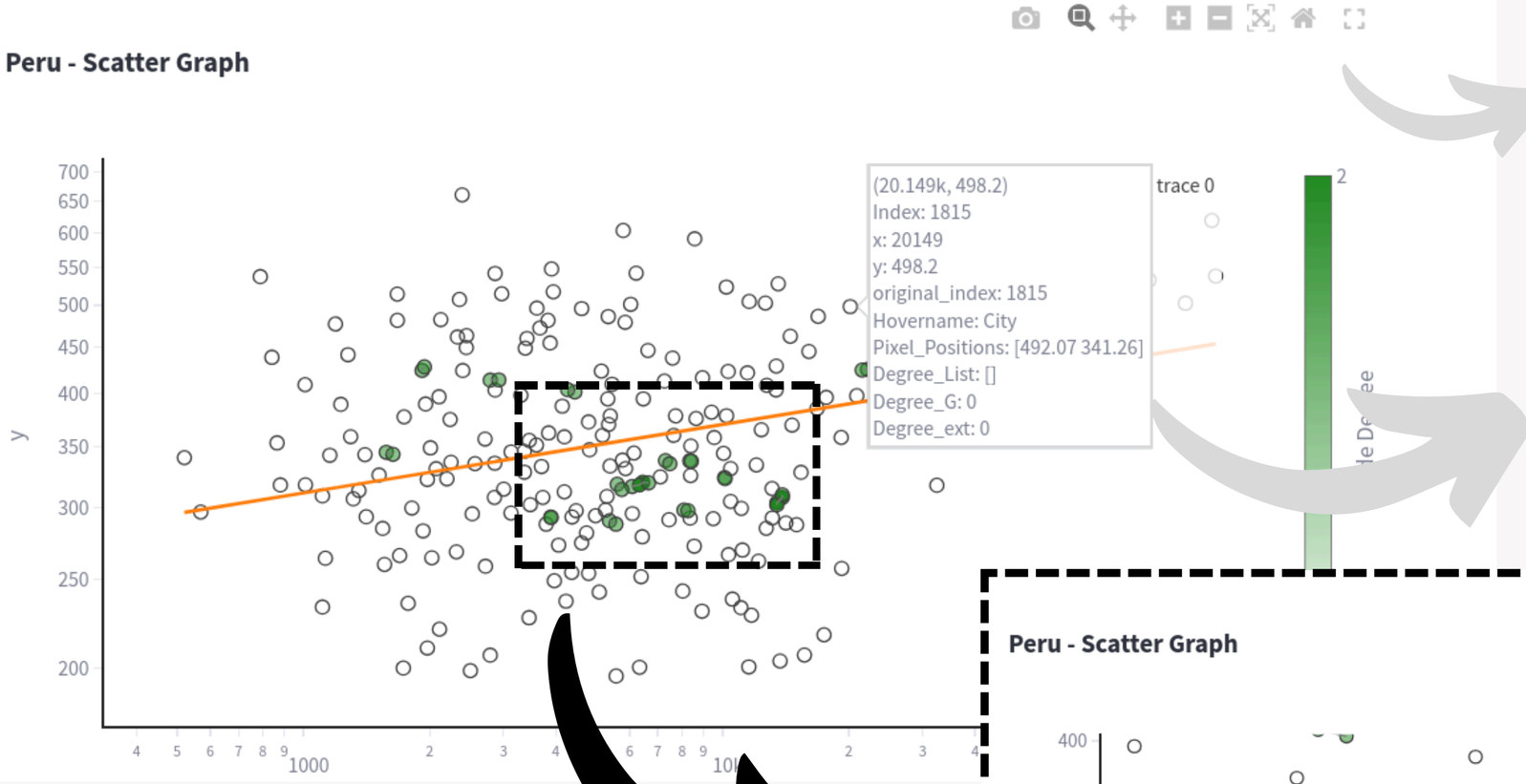
Select Cut Radius Circle (Arbitrary Value)

5

1 100

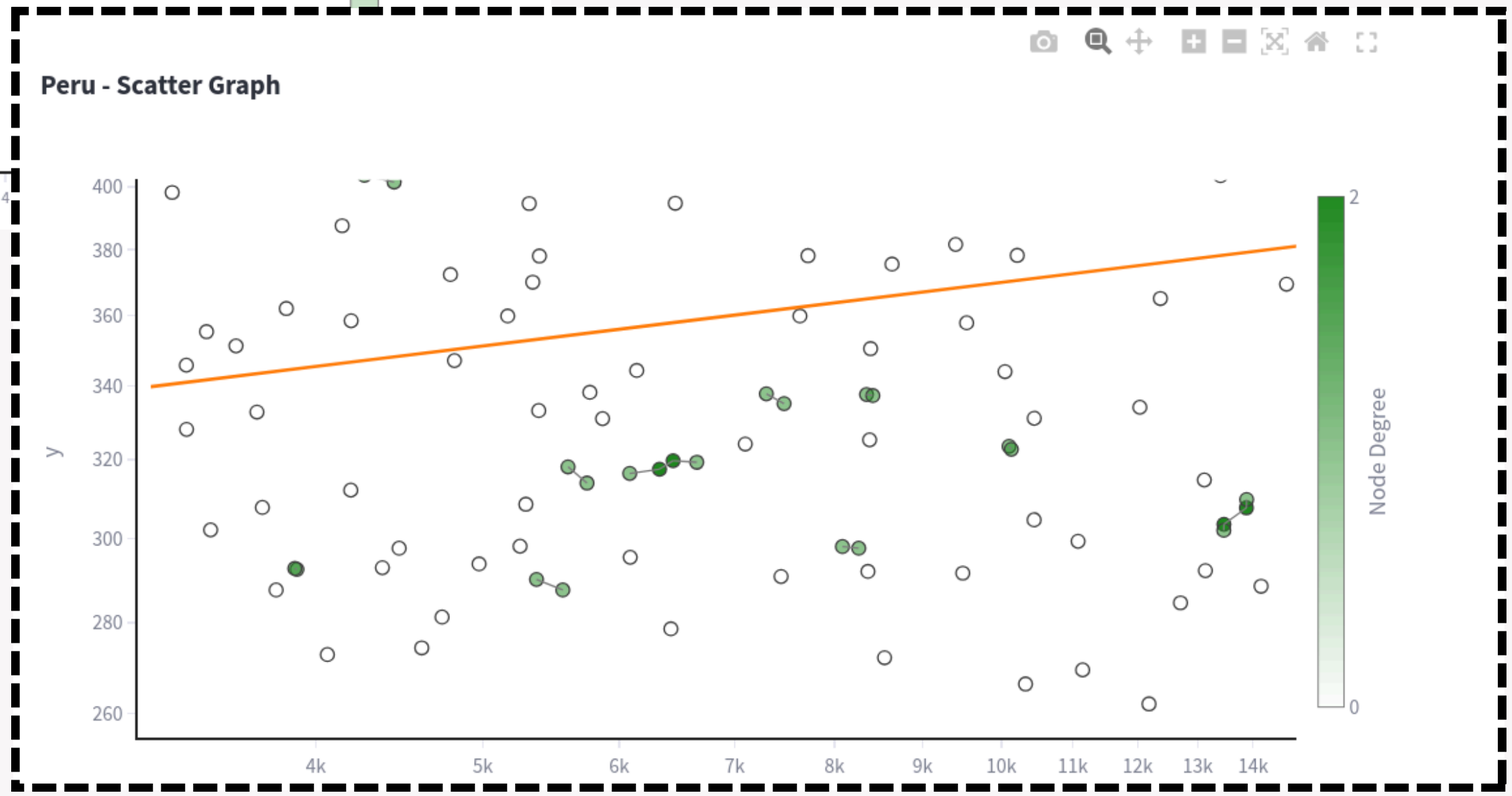
Peru - Scatter Graph





Interactive Menu
Plotly Tools

HoverInfos



Select Box for Zoom In

Scatter which is also a Graph